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Addressing Stuffing Box Leaks at Daleel Petroleum-Oman

Said Al Badi, Omran Al Faraji, Ahmed Nasser Al Abri, Nasser Al Hinai, Mohamed Al Shuely, Ahmed Mohammed Al Naabi, Fahad Al Rahbi, and Abduladhim Al Mandhari, Daleel Petroleum, Oman; Atul Khemnar, Nadhira Enterprise LLC; Hamed AL Qasmi and Maayiya Al Siyabi, Daleel Petroleum, Oman

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Abstract

The operator operates predominantly with sucker rod pumps in their northern Oman field, characterized by diverse reservoir conditions, including high Gas-Oil Ratio (GOR) depleted reservoirs, high water cut, and hydrogen sulphide (H₂S) wells. These conditions have posed significant challenges, particularly with rapid stuffing box (SB) leaks, leading to elevated maintenance costs, increased oil deferment, and significant environmental impacts due to H₂S release and oil slippage. This paper explores the methodologies employed to address the stuffing box leak issue, detailing practical and cost-effective solutions with a focus on environmental protection and H₂S containment.

The challenge of stuffing box leaks not only results in oil losses but also raises serious environmental concerns due to the release of H₂S, a hazardous gas, and the risk of oil contamination. To address these issues, a comprehensive investigation was conducted to identify root causes, encompassing surface beam pump completion and reservoir elements. The investigation facilitated integrated solutions to maintain continuous SB lubrication by optimizing well flowing pressure, enhancing installation and maintenance procedures, and implementing specialty stuffing boxes. In response, various measures were trialed, including a stuffing box leak detector, a polished rod lubricator, and a self-lubricating stuffing box, along with improved polished rod specifications.

The combined cost-effective solutions led to a substantial improvement in stuffing box and polished rod performance over a two-year period. The number of stuffing box leaks was significantly reduced by 86%, and corrective SB maintenance dropped by 42%, even with an increased number of beam pump wells. Consequently, the environmental impact and health and safety risks to personnel were significantly mitigated, primarily through the prevention of H₂S release and reduction in oil slippage. Moreover, the reduction in oil deferment and optimized resource utilization for maintenance teams highlight the effectiveness of the implemented solutions.

This paper presents an in-depth investigation of stuffing box leaks under diverse reservoir conditions, offering integrated, cost-effective, and proven solutions based on data from over 400 beam pump wells over a two-year period, with a strong emphasis on environmental protection and H₂S containment.

INTRODUCTION

Stuffing box leaks are one of the most persistent and costly challenges in beam pumping systems, particularly in oil fields where harsh operating conditions are prevalent. At The operator, stuffing box failures have been identified as a critical bottleneck in maintaining production efficiency, equipment reliability, and environmental compliance. The stuffing box, a key component in the artificial lift system, is responsible for sealing the wellhead while allowing the polished rod to reciprocate smoothly during pumping operations. However, the high mechanical stress and exposure to aggressive well fluids, lack of lubrication make it prone to wear in SB and/or packings, resulting in leaks over time.

The impacts of stuffing box leaks are multifaceted. From an operational perspective, leaks often lead to unplanned downtime, production losses, and increased repair costs. Environmental consequences are also significant, as hydrocarbon leakage contributes to environmental contamination and poses safety hazards. Moreover, repeated failures create additional strain on maintenance teams, stretching resources and increasing operational costs. For The operator, these issues are particularly pronounced in wells with challenging production conditions, such as high gas-oil ratios, high WC% and frequent pressure fluctuations. These factors exacerbate the wear and tear on stuffing boxes and packings, leading to recurring failures that disrupt production operations.

Historically, traditional mitigation methods, such as periodic replacements and ad-hoc repairs, have proven insufficient to address the root causes of stuffing box leaks effectively. The operator recognized the need for a more systematic and proactive approach to improve stuffing box reliability and ensure long-term sustainability in its operations. Key challenges included:

1. High-Pressure Wells: Wells operating under elevated pressure conditions subjected stuffing boxes to excessive mechanical and thermal stresses, accelerating their wear and reducing their lifespan.
2. Gassy Wells: High gas content in produced fluids created operational challenges such as gas locking and insufficient lubrication of stuffing box packing, increasing the likelihood of leaks.
3. H₂S Corrosion: Wells with high H₂S content faced significant corrosion issues, which not only degraded stuffing box and Polished Rod materials but also posed safety risks to personnel and equipment.
4. Material Limitations: Conventional stuffing box materials and designs were not adequately equipped to withstand the combination of mechanical, chemical, and thermal stresses present in the field.

To overcome these challenges, the operator adopted a holistic approach to address stuffing box leaks, combining operational improvements, equipment upgrades, and enhanced surveillance practices. A detailed evaluation of failure modes was conducted to identify key areas of improvement. The following initiatives were prioritized:

- Optimizing the operation of beam pumping systems to prevent gas locking and ensure sufficient lubrication for stuffing box packing.
- Upgrading stuffing box designs and materials to withstand high-pressure and high-H₂S environments.
- Proactively replacing worn-out components such as polished rods and stuffing boxes before they fail.
- Introducing a dedicated surveillance team to monitor well performance and detect early signs of stuffing box wear or failure.
- Conducting field trials with advanced equipment, such as H₂S-resistant stuffing boxes and polished rod lubricators, to evaluate their effectiveness in mitigating leaks.

This paper presents the systematic methodology and strategies implemented at the operator to reduce stuffing box leaks, improve equipment reliability, and enhance production efficiency. By addressing the root

causes and leveraging innovative solutions, the operator has achieved significant reductions in downtime and operational costs while ensuring safer and more environmentally friendly operations. This case study serves as a benchmark for tackling similar challenges in beam pumping systems across the oil and gas industry.

Background

Beam pumping systems, also known as sucker rod pumps, are one of the most widely used artificial lift methods in the oil and gas industry. These systems are favored for their simplicity, cost-effectiveness, and ability to operate efficiently across a wide range of well conditions, from shallow to moderately deep wells. A critical component of these systems is the stuffing box, which plays a pivotal role in sealing the wellhead while allowing the polished rod to move reciprocally. This seal prevents the escape of produced fluids and gases, ensuring the integrity of the system and the safety of operations.

Despite its essential role, the stuffing box is prone to operational challenges due to the harsh conditions encountered in oil wells. High temperatures, pressure fluctuations, abrasive particles, corrosive fluids, and gas production contribute to the wear and degradation of the stuffing box. These conditions can lead to leaks, resulting in environmental, operational, and economic consequences. Common issues associated with stuffing box failures include:

1. **Production Losses:** Leaks can disrupt the artificial lift process, causing reduced efficiency and unplanned downtime.
2. **Environmental Risks:** Escaping hydrocarbons pose significant risks to the environment, increasing the potential for spills and regulatory non-compliance.
3. **Maintenance Costs:** Frequent repairs and replacements of stuffing boxes lead to increased operational expenditures and strain on maintenance resources.
4. **Safety Concerns:** Leaks in wells with high H₂S concentrations can pose serious safety risks to personnel, requiring immediate attention and mitigation.

The occurrence of stuffing box leaks is particularly common in wells producing high gas-oil ratios, H₂S, or abrasive sand and solids, as these factors accelerate the degradation of sealing components. Additionally, improper lubrication, suboptimal operating conditions, and outdated equipment designs can further exacerbate the issue.

To address these challenges, the oil and gas industry has focused on improving stuffing box performance through advancements in material science, equipment design, and operational practices. Modern stuffing boxes are now designed to handle higher pressures, resist corrosion, and improve the longevity of sealing components. The use of lubricators, advanced packings, and corrosion-resistant alloys has significantly reduced the frequency of leaks in challenging environments.

However, solving stuffing box issues is not solely reliant on equipment upgrades. A holistic approach that integrates proactive maintenance, real-time monitoring, and optimized well operations is essential to ensure the long-term reliability of beam pumping systems. Companies that adopt a combination of advanced technologies and systematic practices can minimize stuffing box failures, reduce downtime, and enhance overall production efficiency.

Methods

To address stuffing box leaks and improve beam pumping system performance, a systematic approach was implemented, combining operational optimizations and material upgrades. A reliability study identified root causes and guided the following steps:

1. Identification and Prioritization of Problematic Wells

Wells with frequent subsurface sucker rod (SB) leaks were identified, categorized, and prioritized for intervention.

2. Operational Parameter Optimization

Adjustments to pump strokes, speeds, and fluid levels were made to prevent dry running and reduce mechanical stress.

3. Material Upgrades

A ranked list of problematic wells guided the installation of upgraded materials during workovers, ensuring long-term reliability and performance.

Stuffing Box Upgrades

1. **Diverse Options Deployment:** Four stuffing box types were introduced based on field-specific requirements and new contractual agreements.
2. **High-Performance Upgrades:** Advanced stuffing boxes were installed in high-pressure and problematic wells to mitigate leaks effectively and enhance overall reliability.
3. **H2S-Resistant Solutions:** H2S-resistant stuffing boxes were implemented in wells located in high H2S-producing zones to ensure safety and operational efficiency. Figures 1-5 are showing different types of stuffing box for different applications as mentioned above.



Figure 1-5—different types of stuffing box for different applications

Polished Rod Upgrades

4. **Material and Coating Enhancement:** The specifications for polished rods were upgraded to meet API standards and align with maintenance team recommendations. This included improved material and advanced coating technologies to enhance durability and performance.
5. **Dimensional Upgrades:** The diameter of polished rods was standardized and upgraded to 1.5 inches for improved load-bearing capacity.
6. **Proactive Replacement Campaigns:** A proactive replacement strategy was employed to systematically replace damaged polished rods during routine maintenance, minimizing unplanned failures.
7. **Polished Rod Lubrication:** Trials were conducted to evaluate the performance of lubricators on polished rods, targeting wells prone to gas leaks. These trials demonstrated improved lubrication efficiency and reduced wear.

Optimizing the running modes of the beam pump wells of low flow rate wells to avoid running dry condition

1. **Adjustment of Operating Modes:** The running modes (Pump Fillage Control and Pump Intake Pressure Control mode) of beam pump wells were optimized to prevent lost flow conditions, ensuring consistent and efficient operation.
2. **Gassy Well Management:** Wells producing gas were operated on a predefined Pump Fillage (PF %) mode, with set points optimized to maintain continuous flow and ensure adequate lubrication of stuffing box packing.

Monitoring and Surveillance

3. **Dedicated Surveillance Team:** A specialized LLF (Look Listen and Feel) team was formed to perform daily inspections of beam pump wells. This team focused on identifying defects in stuffing boxes and polished rods, enabling immediate corrective actions and reducing downtime.

Polished rod Lubricator trial

A polished rod lubricator as shown in [figure 7](#), is a specialized device installed above the stuffing box on a sucker rod pumping system to minimize leaks and extend the life of the stuffing box components. It provides continuous lubrication to the polished rod, reducing friction and wear as the rod moves through the stuffing box, enhancing seal integrity and minimizing fluid or gas escape. At The operator, several trials were conducted to evaluate the effectiveness of this technology, demonstrating significant reductions in stuffing box leaks and improvements in system reliability, contributing to enhanced operational performance and reduced maintenance costs.

Stuffing Box Leak Detector trial

A stuffing box leak detector as shown in [figure 6](#), is an innovative device designed to monitor and detect leaks from the stuffing box in sucker rod pumping systems. This tool works by identifying changes in pressure, fluid levels, or flow around the stuffing box, providing early detection of leaks before they escalate into significant issues. At The operator, trials of the stuffing box leak detector were conducted, showcasing its effectiveness in promptly identifying leaks. This proactive detection enabled timely maintenance, reduced downtime, and minimized environmental risks, improving operational efficiency and safety.



Figure 6—Stuffing box leak detector.



Figure 7—Polished rod lubricator.

This integrated methodology combined operational adjustments, continuous monitoring, and hardware enhancements to systematically reduce stuffing box leaks and improve the reliability and productivity of beam pumping systems. Each step was validated through field trials and monitoring to ensure optimal performance.

Discussion of Results

The results clearly indicate a successful multi-phased intervention strategy to mitigate frequent stuffing box packing failures in mature rod-pumped wells operating under challenging conditions such as high H₂S, elevated GOR, and high pump intake pressures.

1. Shift from Reactive to Proactive Packing Strategy

Table 2 illustrates a major transformation from reactive to proactive maintenance between 2021 and 2023. Proactive replacements increased from just 18% in 2021 to 44% in 2023, while corrective actions remained relatively constant. This shift significantly improved early failure mitigation and reduced the frequency of environmental incidents related to oil leaks. The proactive approach allowed better scheduling of maintenance and minimized unscheduled deferment.

In 2024, although there is a slight reduction in proactive actions (33%), this is attributed to the stabilization of failure trends due to the cumulative effect of previous interventions—namely, upgraded stuffing boxes, rod replacement, and continuous surveillance.

2. Proactive Polished Rod Replacement Campaigns

As shown in Figure 8, failures linked to poor polished rod conditions showed a sharp decline after the proactive polished rod campaign began in 2021. The percentage of rods with diameters upgraded to high-tungsten-coated 1½" polished rods significantly contributed to performance improvement, reducing wear-induced failures by nearly 50%. The data also shows a consistent upward trend in rod integrity over time.

3. Stuffing Box Performance and Design Upgrades

Figure 9 demonstrates the comparative performance of different stuffing box types. Upgraded boxes showed substantial improvement in leak rates and packing longevity compared to older designs. Notably:

- Type (1) achieved the best results, with failure rates reduced by 77%.
- Type (4), which showed the poorest performance, was phased out in 2023.

- Introduction of H₂S-specific upgraded boxes (16% failure reduction) reflects a tailored design response to harsh gas environments.

The result validates the importance of matching stuffing box type and metallurgy with downhole conditions.

4. Impact of Monitoring and Surveillance

The 2023 introduction of a Dedicated Surveillance Team enabled physical inspections of 520 wells within 4–6 days. This significantly curtailed the latency between leak occurrence and intervention. Early detection through this program avoided prolonged exposure to surface oil leaks and enhanced HSE compliance. The surveillance system emerged as a key enabler for sustaining the benefits of the proactive replacement strategy.

5. Polished Rod Lubricator Trial

In 2023, a pilot trial involving the installation of polished rod lubricators in two high-GOR wells (Table 3 and 4) demonstrated:

- Packing failure dropped from 6 to 4 and 5 to 0 in the two wells respectively.
- However, the trade-off involved increased cost due to oil top-up (22–23 liters/month) and frequent manual intervention.
- The trial showed a 100% runtime in Well-1 but only 83% in Well-2, reflecting operational variability.
- Fillage improvements (up to 43.57%) and reduced water cut supported the case for improved mechanical efficiency.

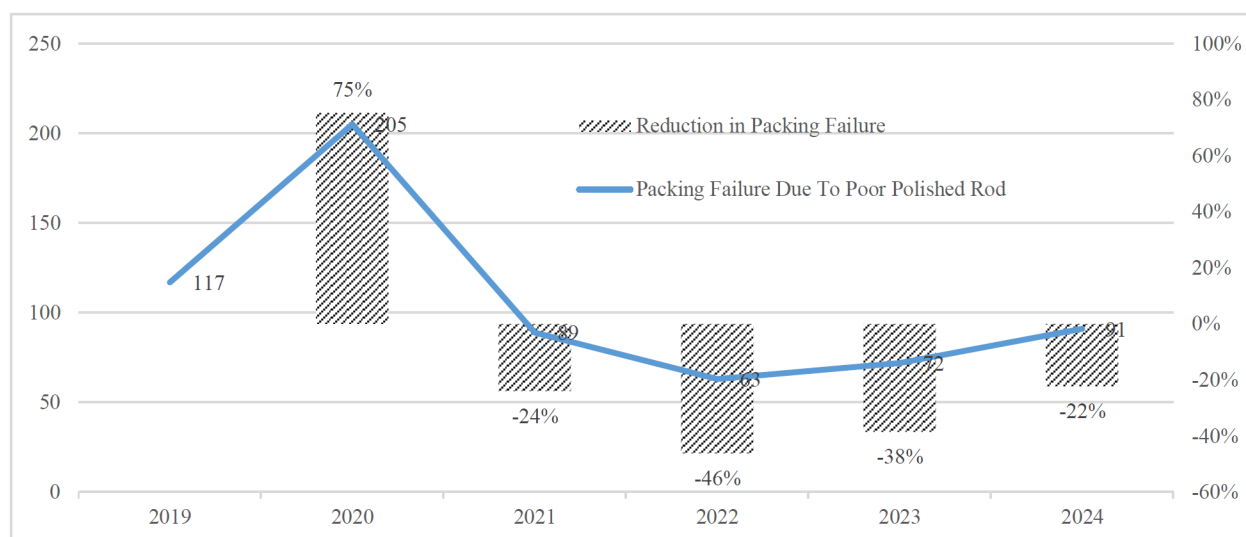


Figure 8—Packing Failure due To Poor Polished Rod

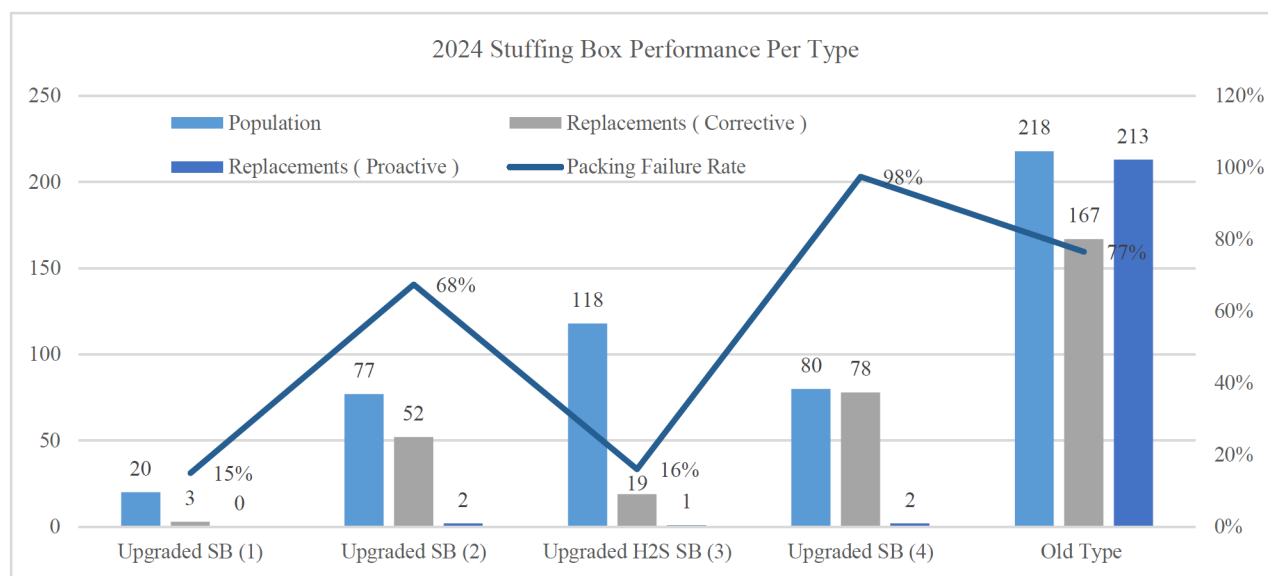


Figure 9—Stuffing Box Performance per Type (2024)

Although the results suggest a marginal packing performance improvement, the lubricator trial may offer incremental value where rod wear is persistent, especially in high-load wells. Cost-benefit analysis is recommended before scaling up this solution.

Table 1—Action implemented to control oil

Action	Effective Date
Stuffing Box Leak Detector trial	2020
Proactive Packing Replacement	2021
Proactive Polished Rod Replacement Campaigns	2021
Stuffing Box Upgrades	2021
Polished Rod Upgrades	2023
Monitoring and Surveillance	2023
Polished rod Lubricator trial	2023

Table 2—Proactive Packing Statistics

Year	Stuffing Box packing Replacement		
	Proactive	Corrective	Proactive percentage
2021	97	441	18%
2022	277	415	40%
2023	299	414	42%
2024	102	205	33%

Table 3—Polished Rod Lubricator Performance

Well	Packing failure 2022 (Without Lubricator)	Packing failure 2023 (with Lubricator)	Packing Change Difference	Packing failure 2024 (with Lubricator)	Packing Change Difference	Lubrication oil top up	Trail Run life
Well – 1 (Old Type)	6	5	1	4	2	23	1-2 Year
Well – 2 (Old Type/ SB type 4)	5	4	1	NA	NA	12	

Table 4—Production of Selected Wells for Polished Rod Lubricator

Well	Production, bbl/d	WC%	Net Oil, bbl/d	GOR, SCF/BBL	Gas Rate, SCF	Avg PF, %	Avg RT, %
Well-1	151.49	98.82	1.79	116245.80	208.08	42.57%	100.00%
Well- 2	89.79	4.19	86.03	1823.30	156.86	37.31%	83.00%

Conclusion

The implementation of a structured proactive maintenance strategy significantly reduced packing failures and associated oil leaks in the operator's mature rod-pumped wells. The transition from a reactive "run-to-fail" approach to a proactive stuffing box packing replacement program—first annually and then semi-annually—yielded measurable benefits. Proactive replacements increased from 18% in 2021 to 44% in 2023, effectively controlling the escalation of failure rates.

Complementary initiatives, including proactive polished rod replacement campaigns, stuffing box design upgrades, and the deployment of a dedicated surveillance team, further enhanced system integrity and early failure detection. The introduction of a polished rod lubrication system showed promise, with selected wells exhibiting improved pump fillage and reduced water cut, despite marginal changes in failure rates.

Collectively, these actions contributed to a robust and scalable strategy that not only minimized environmental impact and operational downtime but also enhanced production reliability. The approach demonstrates a viable model for extending equipment life and improving operational efficiency in high-GOR, high-H₂S, and high-fluid-rate environments typically seen in mature assets.

Recommendations

1. Institutionalize the 6-Month Proactive Packing Replacement Strategy

The shift from a reactive to a biannual proactive stuffing box replacement schedule proved effective in minimizing failure rates and oil leaks. It is recommended to formalize this approach as standard practice across all high-risk rod-pumped wells, particularly in fields with high GOR and H₂S exposure.

2. Standardize Use of Upgraded Stuffing Boxes

The performance of stuffing box type (1) demonstrated superior sealing integrity and resilience under elevated pressure and sour service conditions. Standardizing this design across the asset base is advised, particularly for wells with pressures exceeding 1500 psi and fluid rates above 2000 BFPD.

3. Continue Polished Rod Replacement and Upgrade Campaigns

Proactive polished rod replacements effectively reduced failures attributed to rod wear. The campaign should be maintained annually with consideration to expand the use of high-tungsten 1½" rods, especially for high-stroke and long-stroke units.

4. Enhance Surveillance Capability

The deployment of a dedicated surveillance team significantly shortened detection-response time. Expanding surveillance to include sensor-based diagnostics (e.g., vibration or pressure transients) can further reduce leak risks and support predictive maintenance.

5. Evaluate Cost-Effectiveness of Lubricator Systems

Although polished rod lubricators showed promise in reducing packing wear, the associated operational cost and manpower demand require careful evaluation. It is recommended to limit deployment to high-value, high-maintenance wells until further automation or cost reduction is achieved.

Lessons Learned

1. Proactive Maintenance Prevents Escalation

The introduction of structured preventive actions disrupted the failure cycle, proving that early intervention is significantly more cost-effective and environmentally sound than reactive repairs.

2. One-Size-Fits-All Design Does Not Work

Field results emphasized the importance of selecting stuffing box and rod designs suited to specific well conditions. Matching metallurgy, pressure ratings, and fluid composition to component specifications is critical for long-term integrity.

3. Surveillance and Speed of Response Are Key

Even the best hardware upgrades can fall short without timely monitoring. The ability to physically inspect over 500 wells within a 6-day window played a vital role in maintaining operational integrity.

4. Trial and Data-Driven Iteration Works

Implementing small-scale trials (e.g., rod lubricator, rod upgrade types) and scaling based on field performance data allowed informed decision-making and avoided blanket investments with uncertain payback.

5. Cross-Disciplinary Coordination Is Essential

Success required collaboration across production, maintenance, HSE, and procurement teams. Coordinated execution ensured alignment of surveillance, equipment readiness, and resource mobilization.

Abbreviations and Glossary

SB	Stuffing Box – A surface sealing device used in rod pump systems to prevent fluid leakage around the polished rod.
PR	Polished Rod – The uppermost section of the sucker rod string that passes through the stuffing box and connects to the pumping unit.
GOR	Gas-Oil Ratio – The volume of gas produced per unit of oil, typically expressed in SCF/BBL (standard cubic feet per barrel).
H ₂ S	Hydrogen Sulfide – A toxic and corrosive gas commonly found in sour oil and gas reservoirs.
WC%	Water Cut Percentage – The ratio of water produced compared to the total liquid production, expressed as a percentage.
LLF	Look, Listen, and Feel – A manual surveillance technique used by field teams to identify mechanical issues.
PF%	Pump Fillage Percentage – The ratio of fluid volume displaced by the pump compared to its maximum capacity per stroke.
RT%	Run Time Percentage – The percentage of time a well remains in pumping operation during a given period.

- API American Petroleum Institute – A standards organization whose specifications guide equipment quality and performance in the oil and gas industry.
- SCF Standard Cubic Feet – A unit of gas volume at standard conditions of pressure and temperature.
- bb1 Barrel – A standard unit of volume used in the oil industry, equivalent to 42 U.S. gallons.